Suggested Teaching Guideline For
Concept of Operating Systems & Administration
PGCP- ITISS February 2026

Duration: 90 Theory hrs + 90 Lab hrs + 30 Self-Learning hrs (Total: 210 Hrs)

Objective: To introduce the student to concepts of Operating Systems & Administration.

Evaluation method: Theory exam – 40% weightage
Lab exam – 40% weightage
Internal Assessment – 20% weightage

List of Books / Other training material

Courseware: Linux All-In-One for Dummies, by Emmett Dulaney (Wiley), 6th Ed

Reference book:
- Mastering Windows Server 2016 R2 by Brian Svidergol, Vladimir Meloski
- Windows Server 2022 Administration Fundamentals by Bekim Dauti

Note: Encourage candidates to implement lab using scripts.

Windows Operating System and Security Issue (40 Hrs Theory + 40 Hrs Lab + 15 Self-Learning - 95 Hrs)

Session 1 & 2: (4T + 4L)
Theory:
- Overview of windows operating system
- Installation of windows operating system

Lab:
- Install, upgrade, and migrate servers and workloads
- Create, manage, and maintain images for deployment

Session 3: (2T+ 2L+ 3SL)
Theory:
- Concept of Windows Server Backup (WSB)

Lab:
- Implement Windows Server Backup (WSB)

Self-Learning: (Note: This SL is based on session 1- 3)
- Learn about Windows OS architecture (kernel mode vs User mode)
- Difference between Windows Server editions (Std vs Datacenter)
- Diffrence between In-place upgrade vs clean installation vs migration
- Backup types in WSB (Full, Incremental, System State)
- Case Study: Ransomeware attack on a windows seerve due to missing backups
  Analyze:
  - What went wrong

Suggested Teaching Guideline For
Concept of Operating Systems & Administration
PGCP- ITISS February 2026

- How system state backup could help recover AD

Session 4 & 5: (4T+ 4L)
Theory:

- Implementation, planning and maintaining of active directory infrastructure

Lab:

- Implementing and administering Active Directory
- User accounts and groups in an Active Directory Domain
- Implementing an Active Directory Domain Services (ADDS)
- Domain Controller (DC)
- Additional Domain Controller (ADC)
- Client (Windows 10/11)
- Member Server (MS)

Session 6 & 7: (4T+ 4L+ 3SL)
Theory:

- Configuring DNS
- Implement DHCP and IPAM

Lab:

- Configuration of DNS server
- Install and configure DHCP
- Manage and maintain DHCP
- Implement and Maintain IP Address Management (IPAM)

Self-Learning (Note: This SL session is based on Session 4-7)

- Find out the Logical vs Physical AD components
- Find out the Role of DNS in AD authentication
- Write down Common AD misconfigurations leading to security breaches
- Look into DHCP attacks (Rogue DHCP, DHCP starvation)
- Case Study: Active Directory Compromise due to weak DNS and excessive privileges.

  Analyze:
  - How attackers escalate privileges
  - Importance of least privilege
  - authoritative Vs non-authoritative

Session 8: (2T+ 2L)
Theory:

- Local policies and Group Policies

Lab:

- Implement Local policies & Group policies

Suggested Teaching Guideline For
Concept of Operating Systems & Administration
PGCP- ITISS February 2026

Session 9: (2T+ 2L)
Theory:
- Implement Network Connectivity and Remote Access Solutions

Lab:
- Implement network connectivity solutions
- Implement virtual private network (VPN) and Direct Access solutions

Session 10: (2T+ 2L)
Theory:
- Concept of IIS Web Server

Lab:
- Implement of IIS Web Server

Session 11: (2T+ 2L+3SL)
Theory:
- Concept of Core and Distributed Network Solutions

Lab:
- Implement IPv4 and IPv6 addressing
- Implement Distributed File System (DFS) and Branch Office solutions

Self-Learning: (Note: SL Based session 8-11)
- Learn about IIS attack surface (directory browsing, weak authentication)
- Find out the DFS role in availability and branch office resilience
- Make a comparison on default domain policy Vs custom GPO
- Case Study: Data leakage due to weak Group Policy and IIS misconfiguration

  Analyze:
  - Which GPO settings could have prevented it
  - Which IIS hardening steps could have prevented it

Session 12: (2T+ 2L)
Theory:
- Concept of Windows Deployment Service (WDS)

Lab:
- Deploying Windows 10/11 using WDS

Session 13 & 14: (4T+ 4L)
Theory:
- Implement Hyper-V
- Configure disks and volumes
- Implement server storage
- Implement data deduplication

Lab:
- Install and configure Hyper-V
- Configure virtual machine (VM) setting

Suggested Teaching Guideline For
Concept of Operating Systems & Administration
PGCP- ITISS February 2026

Session 15: (2T+ 2L)
Theory:
- Concept of File Server Resource Manager (FSRM)

Lab:
- Implement File Server Resource Manager (FSRM)

Session 16: (2T+ 2L+ 3SL)
Theory:
- Concept of Network Policy Server (NPS)

Lab:
- Implement Network Policy Server (NPS)

Self-Learning: (Note SL based on session 12-16)
- Make a note of risks in unattended OS deployment
- List out Hyper-V security concerns
- How to use FSRM for ransomware protection
- Case Study: Unauthorized access due to weak NPS policis and poor Vm isolation

  Analyze:
  - How NPS policies should be designed
  - Best practices for Hype-V

Session 17: (2T+ 2L)
Theory:
- Concept of Network Load Balancing (NLB)
- Troubleshooting

Lab:
- Implement of Network Load Balancing (NLB)
- Maintenance and troubleshooting
- Introduction to Microsoft Windows 10 & 11 security
- Security issues at the Active Directory level
- Evaluating and analyzing workstation security
- Securing Windows services

Session 18: (2T+ 2L)
Theory:
- Concept of Exchange server

Lab:
- Installing and implementing of Exchange Server.

Session 19 & 20: (4T+ 4L+ 3SL)
Theory:
- Power Shell

Lab:
- Windows Power Shell Technology Background and Overview

Suggested Teaching Guideline For
Concept of Operating Systems & Administration
PGCP- ITISS February 2026

- Power shell Error Handling and Debugging
- Windows Administration using power shell
- Background Jobs and Remote Administration
- Windows Power Shell Tips & Re-Using Scripts and Functions

Self-Learning: (Note: SL is based on session 17-20)

- Find out about the common windows service vulnerabilities
- What are attack vectors for exchange server
- Explore the event viewer security logs
- Case Study: Exchange server breach via unpatched services
  Analyze:
    - Why patch management failed
    - How powershell logging could help in detection

Linux Operating System and Security (115Hrs) (50 Hrs Theory + 50 Hrs Lab + 15 Hrs Self Learning)

Session 1 & 2: (4T+ 4L)
Theory:

- Introduction to Linux
- The Linux File System
- Working with Files and Directories
- Getting Started to Linux
- Revision of basic Commands
- Access control list and chmod command
- chown and commands
- Network Commands like telnet, ftp, ssh, and sftp,finger
- Use of secondary storage devices (Like: - Hard disk, Floppy, CDROM) in Linux environment and formatting of these devices.

Lab:

- Getting Acquainted with the Linux Environment
- Use various commands in Linux system.
- (ls, cp, mv, lpr, sort, grep, cat, tac, more, head, tail, man, whatis, whereis, locate, find, diff, file, rm, mkdir, rmdir, cd, pwd, ln and ln –s, gzip and gunzip, zip and unzip,tar and its variants, zcat, cal, bc and bc –l, banner date, time, wc, touch, echo, who, finger, w, whoami, who am i, alias, unalias, touch, push, pop, jobs, ps, etc.)

Session 3 & 4: (4T+ 4L+ 3SL)
Theory:

- Installation of Linux
- The interactive Anaconda installer
- A hands-free method of installation
- Understanding the boot procedure
- Configuring the GRUB boot loader

- The Initial RAM Disk
- Understanding run levels
- Repository & Package Management (RPM & DEB)

Lab:

- Installation of Linux, configuring Boot Loader, Yellow dog updater configuration

Self-Learning: (Note: SL based on Session 1-4)

- Learn the differnce between File permission Vs ACLs
- List out the difference between chmod, chown, setfacl
- Learn about the Boot process
- Explore GRUB configuration files
- Case Study: Privilege Escalation Due to Incorrect File Permissions

  Analyze:
  - How world-writable files can be exploited
  - Impact of misconfigured /etc/passwd and /etc/shadow

Session 5: (2T+ 2L)
Theory:

- Shutdown and Installation concepts
- Kick Start Configuration & Customization
- Deployment using Kickstart
- User administration

Lab:

- Installation of basic packages, configuring Kickstart, User Administration

Session 6: (2T+ 2L)
Theory:

- Network address Ipv4/Ipv6
- Using OpenSSH for network communications
- Using VNC for network communication
- Network Authentication

Lab:

- Configuring Network address, OpenSSH, VNC

Session 7: (2T+ 2L+ 3SL)
Theory:

- User & Group Management

Lab:

- User & Group Management

Suggested Teaching Guideline For
Concept of Operating Systems & Administration
PGCP- ITISS February 2026

Self-Learning: (Note: SL is based on session 5-7)

- Find out Kickstart automation risks & its benefits
- Different types of SSH authentication methods (password vs key-based)
- Explore the SSH config (sshd_config) file
- Case Study: Unauthorized Server Access via Weak SSH Configuration
  Analyze:
  - List out the root login risks
  - Find out the Importance of key-based authentication
  - Learn about Password brute-force attacks

Session 8: (2T+ 2L)
Theory:

- Disk management

Lab:

- Fdisk, gdisk, LVM

Session 9: (2T+ 2L)
Theory:

- Network Implementation & print services

Lab:

- Network Implementation & print services

Session 10: (2T+ 2L)
Theory:

- Services Management
- System Configuration Files

Lab:

- Working with System configuration files, NIS Configuration

Session 11: (2T+ 2L+ 3SL)
Theory:

- Patches
- System Management
- X configuration server

Lab:

- Working with patch management, X configuration

Self-Learning: (Note: SL is based on session 8-11)

- Explore about Disk partitioning risks (MBR vs GPT)
- List out LVM advantages in recovery & security
- Explore Patch update workflows
- Explore about Linux service lifecycle (systemctl)
- Case Study: Data Loss Due to Poor Disk & Patch Management
  Analyze:

- How LVM snapshots help recovery
- Risks of unpatched services
- Downtime caused by mismanaged system services
- Analyze the RAID configuration in Linux

Session 12: (2T+ 2L)
Theory:

- Configuration of DNS

Lab:

- Configuration of DNS

Session 13: (2T+ 2L)
Theory:

- Introduction to Configuring Services
- Setting up an NFS server
- Setting up an FTP server

Lab:

- Configuring NFS

Session 14: (2T+ 2L)
Theory:

- The Samba Server: networking with Windows system
- Configuring a DHCP server
- Configuring a DNS server

Lab:

- Configuring Samba, DHCP Server, DNS Server

Session 15: (2T+ 2L)
Theory:

- Configuring the Apache web server
- Apache security & Virtual Hosting
- Configuring the Squid web proxy cache

Lab:

- Configuring Apache Web Server, Virtual hosting, Squid Proxy

Session 16: (2T+ 2L+ 3SL)
Theory:

- Understanding e-mail delivery
- Postfix Mail Server
- Dovecot: an IMAP and POP server
- squirrel Mail Web Client

Lab:

- Configuring Postfix, Dovecot, Squirrel Mail

Suggested Teaching Guideline For
Concept of Operating Systems & Administration
PGCP- ITISS February 2026

Self-Learning (Note: SL is based on session 12-16)

- Find about the DNS, DHCP & Samba attack vectors
- Explore Apache configuration files
- Explore Samba permission mapping
- Explore DNS zone files
- Do a FTP vs SFTP security comparison
- Case Study: Web Server Compromise Due to Weak Apache Configuration

  Analyze:
  - Find the Importance of virtual host isolation
  - List out the Directory listing risks

Session 17: (2T+ 2L)
Theory & Lab

- Introduction to Performance Tuning
- Maintenance and troubleshooting
- The Threat Model and Protection Methods

Session 18: (2T+ 2L)
Theory & Lab

- Basic Service Security
- Logging and NTP
- BIND and DNS Security

Session 19: (2T+ 2L)
Theory:

- Network Authentication: LDAP & NIS
- Apache Clustering
- Load Balancer

Lab:

- Configure and implement of LDAP.
- Configure and implement of NIS.
- Configure NTP server for time synchronization on local network

Session 20: (2T+ 2L)
Theory & Lab

- Virtual Machine management
- Virtual machine Network Configuration

Session 21 & 22: (4T+ 4L)
Theory:

- Introduction to BASH Command Line Interface (CLI) Error Handling
- Debugging & Redirection of scripts

- Control Structure, Loop
- Variable & String
- Conditional Statement Regular Expressions

Lab:

- Working to automate bash commands using bash scripts.
- Bash script using loops, Strings, regular expressions

Session 23: (2T+ 2L)
Theory:

- Automate Task Using Bash Script
- Security patches

Lab:

- Working to automate bash commands using bash scripts.

Session 24: (2T+ 2L)
Theory:

- Logging & Monitoring using script

Lab:

- Working to automate bash commands using bash scripts.

Session 25: (2T+ 2L+ 3SL)
Theory & Lab

- Case Study

Self-Learning: (Note: SL based on session 17-25)

- Explore about Linux threat model
- Learn the importance Logging (/var/log)
- Find out about NTP & time-based attacks
- List out LDAP & NIS security concerns
- Case Study: Delayed Breach Detection Due to Poor Logging & Monitoring.

  Analyze

  - How logs help Forensic analysis
  - How automatin scripts can become attack vectors
  - Explore Log analysis techniques